Maciej Krzysztof Zuziak

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PROFILE

AI researcher specialising in the security and privacy of decentralised machine learning – federated learning, membership inference attacks, differential privacy, and data leakage. Marie Skłodowska-Curie Doctoral Fellow (under 10% acceptance rate) and Max Weber Fellow (under 3% acceptance rate) with 5+ years of experience delivering EU-funded research (€20M+ programmes) across academia and industry. Combines rigorous ML engineering with formal legal training in data rights and compliance – able both to design privacy-preserving systems and to assess their regulatory implications. Experienced in independent research leadership, interdisciplinary collaboration, and translating theory into deployable decentralised systems.

EXPERIENCE

- **Institute of Science Tokyo** *Oct 2026 – Dec 2026*
  Visiting Researcher (Incoming), School of Computing Tokyo, Japan
  - Research on backdoor propagation through the model supply chain.
- **European University Institute (EUI)** *Sep 2026 – Sep 2027*
  Max Weber Fellow, Department of Law Florence, Italy
  - Research fellowship awarded for independent research on **GDPR compliance and privacy considerations regarding Large Language Models**.
  - Selected as **one of 56 distinguished researchers** (out of more than 2000 applications) to conduct independent research under the supervision of senior researchers from the EUI Department of Law.
- **University of Leeds** *Jan 2026 – Jun 2026*
  Research Fellow, School of Computer Science Leeds, United Kingdom
  - **Led independent research** on decentralised systems for Smart Grids, UAV coordination, and Digital Democracy, identifying new methodological directions for the lab.
  - Co-author on *The Optimization Trilemma* (IEEE ACSOS 2026; arXiv:2607.17311), extending decentralised multi-agent coordination to optimise system efficiency, individual comfort and fairness jointly.
  - Surveyed emerging research areas and co-authored **joint grant proposals** targeting Horizon Europe, UKRI, and ARIA funding schemes.
  - Mentored junior researchers and coordinated lab-critical projects, contributing to research continuity and delivery.
- **National Research Council of Italy (CNR)** *Nov 2021 – Dec 2025*
  Research Associate, Knowledge Discovery Lab Pisa, Italy
  - **Core contributor to two Horizon-scale EU projects** (€11M+), focusing on federated personalisation, explainability, and privacy risk mitigation.
  - **Published 6 peer-reviewed outputs** and delivered 1 journal article currently under review.
  - **Designed and implemented reusable ML libraries** for decentralised learning, including controlled non-IID splits, clustered personalisation, and Shapley-based contribution analysis.
  - Contributed methodological components adopted across multiple project work packages and experimental pipelines.
  - Collaborated with leading European research institutions CNRS, Scuola Normale Superiore, Scuola Superiore Sant'Anna, and other European partners on tractable federated algorithms and data governance frameworks.
- **Tellu AS** *Sep 2023 – Sep 2024 (dual appointment)*
  Visiting Research Engineer, DevOps Department Oslo, Norway
  - **Co-developed a production-grade last-mile IoT gateway** with autonomous onboarding and decentralised fleet and device-health management.
  - Designed decentralised Digital Twin architectures for device enrolment, metrics collection, and lifecycle monitoring.
  - Delivered technical documentation for **three Horizon Europe mid-term reviews, all positively evaluated**.
- **Permanent Mission of the Republic of Poland to the United Nations** *Jul 2019 – Aug 2019*
  Intern, Security Council New York, USA
  - Attended Security Council sessions on peacekeeping, counter-terrorism, and transnational crime; engaged with Sanctions Committees (DR Congo, Libya) and **commended by the Head of Mission**.

Education

- **University of Pisa** *Nov 2021 – Dec 2025*
  Ph.D. (Artificial Intelligence) Pisa, Italy
  - Thesis: *Data Collaboration with the Use of Federated Learning: Issues of Marginal Contribution Quantification and Personalisation* (evaluation: eccellente - highest distinction). - Thesis Repository.
  - Joint National Doctorate in AI coordinated with, *among others*, Scuola Superiore Sant'Anna, Scuola Normale Superiore, University of Trento, and University of Bologna.
  - 300+ hours of advanced CS & mathematics modules (graded A–A+).
  - All three annual appraisals rated exceptional.
- **University of Silesia in Katowice** *Sep 2016 – Jul 2021*
  Magister Prawa (Master of Laws, integrated five-year programme) Katowice, Poland
  - Thesis: *Digitalisation of Factoring Contracts* (evaluation: bardzo dobry - highest distinction).
  - Post-graduation prize for most promising graduates.

Journals

[1] **[First]** **M.K. Zuziak**, R. Pellungrini, S. Rinzivillo (2025): **One-Shot Clustering for Federated Learning Under Clustering-Agnostic Assumption**. ACM Transactions on Intelligent Systems and Technology (revision under review) - ArXiv Preprint.

Publications

First = first author; Co-first = equal contribution, as marked in the publication

[1] **[Co-first]** S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning**. *Proceedings on Privacy Enhancing Technologies* 2026(4), pp. 480–506. Presented at *The 26th Privacy Enhancing Technologies Symposium*, Calgary, Canada - PoPETs Access. (*equal contribution)

[2] J. Nikolic, **M.K. Zuziak**, E. Pournaras (2026): **The Optimization Trilemma: Efficiency, Comfort and Fairness in Decentralized Multi-agent Coordination**. At *IEEE ACSOS 2026* — 7th IEEE International Conference on Autonomic Computing and Self-Organizing Systems, Cesena, Italy, 7–11 September 2026. IEEE (to appear in IEEE Xplore) - ArXiv Preprint.

[3] **[Co-first]** S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini, S. Rinzivillo (2025): **Can Contributing More Put You at a Higher Leakage Risk? The Relationship Between Shapley Value and Training Data Leakage Risks in Federated Learning**. At *SECRYPT 2025*, Bilbao, Spain. SciTePress - SciTePress Access. (*equal contribution)

[4] **[First]** **M.K. Zuziak**, R. Pellungrini, S. Rinzivillo (2024): **One-Shot Clustering for Federated Learning**. At *IEEE Big Data 2024*, Washington, USA. IEEE Xplore - IEEE Xplore Access.

[5] **[First]** **M.K. Zuziak**, S. Rinzivillo (2024): **Amplified Contribution Analysis for Federated Learning**. At *IDA 2024*, Stockholm, Sweden. Springer Lecture Notes in Computer Science (LNCS) - Springer Nature Access.

[6] **[Co-first]** **M.K. Zuziak***, O. Hinrichs*, A. Abdrassulova, S. Rinzivillo (2023): **Data Collaboratives with the Use of Decentralised Learning**. At *ACM FAccT 2023*, Chicago, USA. ACM - ACM Digital Library Access. (*equal contribution)

Policy Contributions

[1] ACM ETPC (2026). **Comments in Response to European Commission Call for Evidence Survey on the "Digital Omnibus on AI Regulation" Proposal**. Contributors: G. Besanson, M.K. Zuziak, A. Saucedo, F. Medeiros, T. Romanoff. ACM ETPC Link.

[2] ACM ETPC (2025). **Consultation on the Guidelines on Protection of Minors Online under the Digital Services Act**. Contributors: T. Romanoff, A. Saucedo, M.K. Zuziak, M. Beaudouin-Lafon, C.L. Hankin. ACM ETPC Link.

[3] ACM ETPC (2024). **Comments on Draft Delegated Regulation on Data Access (DSA)**. Contributors: A. Deshpande, L. DiBiasi, M.K. Zuziak, J. Rozanec, A. Nagy, A. Saucedo, F. Medeiros, J. Mizeria-Petraszko. ACM ETPC Link.

Workshops, Posters & Opinions

W=Workshops, P=Posters, O=Opinions

[P.3] S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning (Poster Version)**. Privacy Enhancing Technologies Symposium, Canada. (*equal contribution).

[P.2] **M.K. Zuziak**, S. Rinzivillo (2024): **Personalization via Clustering in Federated Learning for Accessible Digital Services**. School of Advanced Studies Sant'Anna at Pisa Internet Festival, Italy.

[O.1] **M.K. Zuziak** (2024): **How to Collaboratively Use Statistical Models in a Secure Way**. *Opinio Juris in Comparatione*. Opinio Juris Link.

[P.1] **M.K. Zuziak**, S. Rinzivillo (2022): **Federated Learning for Personal Data Management under European Digital Strategy**. Conference on Data Sciences and EU Regulations, IRIT at Université Toulouse III, France.

[W.1] **M.K. Zuziak**, S. Rinzivillo (2022): **Federated Learning as an Analytical Framework for Personal Data Management - a Proposition Paper**. Hybrid Human Artificial Intelligence (HHAI) Workshop, Amsterdam, Netherlands. CEUR Proceedings - CEUR WS Link.

Recent Talks and Seminars

[1] Institute of Science Tokyo (2026): **On the Intrinsic Vulnerabilities in Federated Learning and Corresponding Defence Methods.**

[2] ACM (2026): **Europe's Digital Omnibus: A New Digital Regime or Simplification.** ACM Recording.

Selected Projects

- **Project Tango (Horizon Europe No. 101120763)** *2024 – 2025*
  Research Associate Project page
  - **Total Funding:** €7M
  - **Key Partners:** Université Paris Cité, TU Darmstadt, University of Pisa, Scuola Normale Superiore.
  - **Role:** Research on clustered federated learning and explainability for human-centric decision-making.
- **Project LEADS (MSCA ITN No. 956562)** *2021 – 2024*
  Marie Skłodowska-Curie Actions (MSCA) Doctoral Fellow Project page
  - **Total Funding:** €3.8M
  - **Key Partners:** University of Pisa, Scuola Superiore Sant'Anna, CNRS, University of Luxembourg.
  - **Role:** Research on trustworthy and decentralised ML infrastructure for the Digital European Strategy.
- **Project ENTRUST (Horizon Europe No. 101095634)** *2023 – 2024*
  Visiting Contributing Researcher Project page
  - **Total Funding:** €5.6M
  - **Key Partners:** Tellu AS, TU Eindhoven, Siemens, Sintef.
  - **Role:** Contributed to decentralised trust management architecture; designed risk assessment processes for critical IoT infrastructure (last-mile medical gateways).
- **Project Eratosthenes (Horizon 2020 No. 101020416)** *2023 – 2024*
  Visiting Contributing Researcher Project page
  - **Total Funding:** €6M
  - **Key Partners:** Tellu AS, TU Graz, KU Leuven, Airbus Cybersecurity, Sintef.
  - **Role:** Contributed to an autonomous onboarding flow for critical IoT infrastructure (last-mile medical gateways and sensors connected to them).

Open Source

- **FedJust** *2021–2024*
  Federated Learning toolkit for rapid prototyping and reproducible evaluation of novel federated algorithms. GitHub
- **OCFL Suite** *2024–2025*
  Experimental framework for designing and benchmarking Federated Clustered Learning methods. GitHub

Awards

- **Max Weber Fellowship** *2026*
  European University Institute (under 3% acceptance rate)
- **Horizon Europe (Project Tango) Fellowship** *2024*
  National Research Council of Italy (CNR)
- **Marie Skłodowska-Curie Actions Fellowship** *2021*
  European Commission (under 10% acceptance rate)
- **CSC Scholarship** *2020*
  China Scholarship Council
- **Academic Scholarship (Top 5%)** *2017*
  University of Silesia (awarded multiple times 2017–2020)

Memberships

- **ACM Europe Technology Policy Committee (ETPC)** *Dec 2023 – Present*
  Non-voting member

Research Interests

- Federated and decentralised learning
- Privacy-preserving ML (differential privacy, membership inference, data leakage)
- Evaluation and benchmarking of ML and LLM systems (privacy leakage, agentic behaviour)
- LLM security: backdoor implantation and propagation through fine-tuning and distillation, data poisoning, jailbreaking
- Explainable AI and contribution analysis in distributed settings
- AI governance, liability, and regulation

Skills

- **Core Expertise:** Federated Learning, Privacy-Preserving ML, Explainable AI, Data Governance
- **Programming:** Python, Julia, C/C++, TypeScript
- **ML & Data:** PyTorch, TensorFlow/Keras, JAX, Scikit-Learn, NumPy, Pandas
- **Distributed & Systems:** Flower, PySyft/PyGrid, TensorFlow Federated, Docker, Kubernetes, Linux/WSL
- **HPC:** EuroHPC supercomputers (MeluXina, Leonardo Booster), GPU training
- **Agentic & LLM Systems:** LangChain, LangGraph, RAG pipelines, vector databases (FAISS, Pinecone, Weaviate), LoRA/PEFT

Additional Information

Languages: Polish (Native), English (C2), Chinese (B2), Italian (B1/B2)